

Bridge Weigh-in-Motion as a regularized Bayesian inverse problem: verification, validation against measured data, and model-error-aware uncertainty

Extended abstract — computational engineering portfolio

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Live demo: <https://bypire.github.io/bridgebeat> · Source: github.com/bypire/vbi-bridge-sim (private, access on request)

1. Motivation

Road and bridge damage grows with roughly the **fourth power of axle load** (AASHO Road Test): a truck at twice the legal axle weight does about **16×** the structural damage. **Bridge Weigh-in-Motion (B-WIM)** turns an already-instrumented bridge into a scale that weighs trucks *at traffic speed* by inverting the measured structural response — cheaper and harder to evade than static weigh stations. The same dynamic response also carries information about the structure's *health* (drive-by / vibration-based monitoring). This project builds the **forward** physics (a vehicle crossing a bridge), runs it **backwards** to recover the axle weights as an inverse problem, and validates the result against an analytical reference, a regularization analysis, and **real measured bridge data**. The governing rule: every claim is checkable, and every modelling choice is defensible. The core is numpy-only.

2. Method

The bridge is a **simply-supported Euler–Bernoulli beam** discretised with 2-node beam finite elements (2 DOF/node), giving a consistent mass matrix **M** and bending stiffness **K**; $\mathbf{M} \ddot{\mathbf{u}} + \mathbf{C} \dot{\mathbf{u}} + \mathbf{K} \mathbf{u} = \mathbf{F}(\mathbf{t})$ with optional Rayleigh damping. The **moving load** is mapped to its position each step via cubic Hermite shape functions. The **vehicle** is a 2-DOF **quarter-car** coupled to the beam through the contact force (two-way interaction). Time integration uses explicit **RK4** and implicit **Newmark-β** (the latter unconditionally stable). The reference structure is a 40 m highway overpass ($f_1 \approx 2.6$ Hz, $\zeta = 2\%$). The **inverse** problem recovers static axle loads by fitting the measured bending moment to the section **influence line** (Moses' least squares), extended to a **Tikhonov-regularized** and a **Bayesian** estimator.

3. Verification (the math is correct)

Each building block is checked against an independent **analytical** reference; if the code disagrees, the code is wrong.

Check	Reference	Result
Static mid-span deflection	$P L^3 / 48 E I$	rel. err $\approx 1 \times 10^{-13}$
Natural frequencies	$(n\pi/L)^2 \sqrt{EI/\bar{m}}$	mode 1 $\approx 4 \times 10^{-7}$
Moving constant force	Fryba closed form	peak $\approx 6 \times 10^{-6}$
Quarter-car frequencies	analytic 2×2 eigenproblem	within FFT resolution
Damping ζ	free-vibration log-decrement	within 0.1 %
Newmark-β vs Fryba	closed-form moving force	match to $\approx 2 \times 10^{-4}$
Model order reduction (3 modes)	full-order response	reproduces to 0.16 %

4. Validation and uncertainty (the model means something)

Verification is not validation. Three results separate "the math is right" from "the answer is trustworthy."

4.1 Model-error-aware uncertainty. Casting B-WIM as a linear-Gaussian inverse gives a recovered weight with a 95 % credible interval. A *self-consistent* coverage test (forward and inverse both use the static influence line) returns 95.3 % coverage — but this only verifies the linear-Gaussian machinery. The honest test replaces the forward model with the **full coupled dynamic FEM**: the recovered weight is then biased by dynamic amplification (**+0.3 % quasi-static rising to +5 % at 40 m/s**), and because the residual σ measures goodness-of-fit (the signal still looks influence-line-shaped) rather than the dynamic *scale* error, the nominal-95 % interval's true coverage **collapses to $\approx 0\%$** . The RMS dynamic bias over the highway band ($\approx 3.1\%$) is the model-error term that restores calibration — and it matches the value independently used in the operational dashboard, so that uncertainty is now *calibrated from physics*, not assumed.

4.2 Regularizing the ill-posed axle split. Moses' method recovers **gross** weight to $< 1\%$ but splits **closely spaced (tandem) axles** poorly. This is purely linear-algebraic: a tandem makes two columns of the influence matrix nearly parallel, so $\text{cond}(\mathbf{C}^\top \mathbf{C})$ rises sharply (≈ 9 at 8 m spacing $\rightarrow \approx 2000$ at 0.5 m) and the split amplifies

noise. **Tikhonov regularization** with an **L-curve**-chosen weight cuts the per-axle scatter $\approx 6\times$ ($16\% \rightarrow 2.6\%$) for a small bias, while leaving the well-posed gross weight unchanged — the bias–variance trade-off made explicit.

4.3 Validation against real data (KW51). The 16-month tracked-modal-frequency record of the instrumented KW51 railway bridge (Leuven; public dataset) is ground truth no model can argue with. It confirms the drive-by-SHM premise — a real, dated **retrofit** (stiffening) shifts the modal frequencies by up to **+2.6 %** — and exposes its dominant confounder: **temperature** swings the same frequencies by a comparable amount (**up to $\approx 3.8\%$** , with some modes' temperature swing *exceeding* the retrofit jump). Removing the temperature trend (fit on the healthy period) sharpens the change from a raw to a temperature-corrected residual (e.g. **9.8 σ** $\rightarrow$ **17.3 σ** for one mode). This is exactly the limit the synthetic damage study predicted — frequency is a blunt, environment-contaminated indicator — now confirmed in the field.

4.4 From weight to decision (fatigue economics). A recovered weight is only useful if it drives a decision. Steel-detail fatigue follows an S-N law with slope m (Eurocode EN 1993-1-9, $m = 3$; AASHTO pavement $m = 4$), so damage per pass scales as $(W/W_{\text{ref}})^m$: a **2 $\times$ overload does 8 $\times$ –16 $\times$ the damage** (exact). Over a simulated traffic stream this concentrates sharply — the heaviest **10 %** of trucks cause $\approx$ **45 %** of the fatigue damage and the overloaded minority $\approx$ **half** — and B-WIM identifies exactly those trucks. An asset-depreciation estimate puts one heavy crossing at $\approx$ **€10** of bridge life (vs €0.80 for a legal pass): the quantified case for self-funding overload enforcement.

5. Limitations (honest scope)

One simply-supported single span; linear, small-displacement Euler–Bernoulli theory. A single quarter-car (one wheel path); the multi-axle B-WIM uses constant axle forces. The drive-by study **detects** a global stiffness change but does **not localize** it — a single fundamental frequency cannot, by construction. KW51 is a 115 m steel bowstring **arch**, not a simply-supported beam, so it validates the SHM *behaviour and method*, not the FEM geometry. B-WIM uses the analytical influence line; field systems calibrate it from a known truck and lose accuracy to roughness, multiple vehicles, and temperature.

6. References

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7. COST323, *European Specification on Weigh-in-Motion* — accuracy classes.